Finite Obstruction Calculus: Exact Quotient Diagnostics, Dual Witnesses, Two-Stage Coherence, and Transport (CODA-FOC)

Subtitle: Exact quotient diagnostics, dual witnesses, two-stage coherence, and transport

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Abstract

A defect is a vector that you are allowed to change only by adding a licensed linear repair. What cannot be cancelled is the leftover. This paper measures that leftover in a finite-dimensional weighted ℓ^1 quotient, then measures the image of the same vector under a compatible next-defect map in a second quotient. The two numbers classify every well-posed input into exactly one of three labels: the vector is an exact repair, a coherent residual, or a nonclosed defect. Strong linear-programming duality supplies a dual witness on the whole repair subspace; direct sums add; and a compatible map of carriers cannot inflate leftovers beyond its operator norm. The three labels classify the vector. They do not, by themselves, choose an action.

Introduction

Obstruction language is easy to overstate. In a finite setting the claim "this vector still has a leftover" can be tied to four concrete objects: a declared repair subspace, a strictly positive coordinate weight, an exact linear program, and a dual vector certifying that no cheaper repair exists. The resulting calculus is small enough to replay by hand on a two-dimensional example, and strong enough to catch common errors — substituting least squares for ℓ^1, allowing a zero weight, annihilating only named generators rather than the whole repair subspace, or reading a diagnostic label as a decision.

The mathematical components of this calculus — quotient norms [1, 4], linear programming duality [2, 3], cochain complexes and obstruction cocycles [5, 6], direct sums of complexes [5], and operator-norm bounds [3, 4] — are standard tools in discrete analysis, homological algebra, and polyhedral geometry. The contribution of this paper is their explicit, certificate-producing assembly into an exact two-stage diagnostic and evidence contract that strictly separates mathematical classification from operational authorization.

Figure 1 illustrates three separate finite obstruction packets matching the prose (`public_export/fig1_authority_descent.png`). Take $C_1 = \mathbb{R}^2$ and allow repairs along the first axis: adding any multiple of $(1, 0)$. The vector $(1, 0)$ can be cancelled completely ($d_0 f = (-1, 0)$, residual $r = 0$). The vector $(0, 1)$ cannot; its leftover, in the ordinary ℓ^1 norm, is 1. If the next-defect map is the zero map, that leftover produces nothing at the next stage ($\Phi_1 = 1, \Gamma_2 = 0$). If instead the next-defect map reads the second coordinate ($d_1(a, b) = b$), the same leftover appears one degree up ($\Phi_1 = 1, \Gamma_2 = 1$). Those are the three diagnoses of the paper, in order: exact repair, coherent residual, nonclosed defect.

The implementation or evidence contract is refuted by a valid packet for which a reported quotient value is negative or fails to attain its claimed optimum, a supplied primal and dual certificate fails replay, the reported classifier is not total and exclusive, a claimed annihilator fails on the repair subspace, direct-sum additivity fails, or a compatible transport exceeds its stated bound. The mathematical theorems would be refuted only by a counterexample satisfying their stated hypotheses.

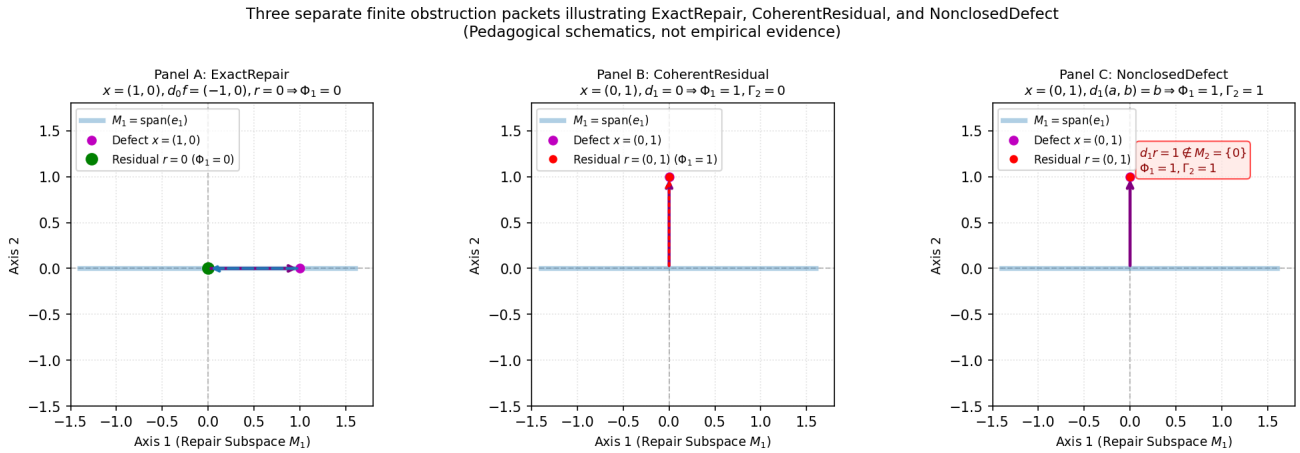


Figure 1: Three separate finite obstruction packets illustrating ExactRepair, CoherentResidual, and NonclosedDefect. The panels are pedagogical schematics, not empirical evidence.

Notation

Throughout, all vector spaces are finite-dimensional and real.

- C_0, C_1, C_2 are the carriers. C_1 is the space of first-stage defects; C_2 is the space of next defects.
- $d_0 \colon C_0 \to C_1$ lists the licensed first-stage repairs. Adding $d_0 f$ to a vector in C_1 is an allowed exact correction.
- $d_1 \colon C_1 \to C_2$ is the next-defect map. The cochain condition $d_1 \circ d_0 = 0$ says that an exact first-stage repair produces no next-stage defect.
- $a_2 \colon U_2 \to C_2$ lists the licensed second-stage repairs of the next defect.
- w_1 and w_2 are strictly positive coordinate weights on C_1 and C_2. They are the observer: they say how expensive each coordinate of leftover is.
- $M_1 = \operatorname{im} d_0$ and $M_2 = \operatorname{im} a_2$ are the full linear repair subspaces, rather than an unclosed list of named generators.
- $Q_1 = C_1/M_1$ and $Q_2 = C_2/M_2$ are the quotients in which leftovers live.
- The weighted ℓ^1 norm is $\|v\|_{1,w} = \sum_i w_i |v_i|$.
- $\Phi_1([x])$ is the cheapest first-stage leftover of x.
- $\Gamma_2([d_1 x])$ is the cheapest second-stage leftover of $d_1 x$.

A **finite obstruction packet** is the tuple $(C_0, C_1, C_2, U_2, d_0, d_1, a_2, w_1, w_2)$ of these spaces, maps, and weights.

1. Finite obstruction packets

A packet consists of linear maps

$$C_0 \xrightarrow{d_0} C_1 \xrightarrow{d_1} C_2, \qquad U_2 \xrightarrow{a_2} C_2$$

with $d_1 d_0 = 0$ and strictly positive weights w_1 on C_1, w_2 on C_2. The nilpotence condition $d_1 d_0 = 0$ defines a two-stage cochain complex in the sense of homological algebra [5, 6]. The matrix of d_0 must span every licensed linear first-stage repair; the matrix of a_2 must span every licensed linear second-stage repair. A list of informal generators is not enough unless its columns already span those images.

Set

$$Q_1 = C_1/M_1, \qquad Q_2 = C_2/M_2,$$

$$\Phi_1([x]) = \min_f \|x + d_0 f\|_{1,w_1},$$

$$\Gamma_2([d_1 x]) = \min_u \|d_1 x + a_2 u\|_{1,w_2}.$$

The second quantity is invariant under first-stage repair, because $d_1(x + d_0 f) = d_1 x$.

Definition FOC-D1 — valid packet

A packet is valid only when all dimensions match, all entries are finite, $d_1 d_0 = 0$ exactly; this release's exact rail accepts exact integer or rational data and does not use numerical tolerances, and every observer weight is strictly positive. Invalid packets are refused before any mathematical classification.

2. Quotient values and dual witnesses

Theorem FOC-T1 — quotient norms and attainment

Φ_1 and the corresponding norm on Q_2 are well-defined attained norms.

Proof. In finite dimension, the repair subspace M is closed. The value at $m = 0$ is $\|x\|_{1,w}$. By the reverse triangle inequality, every $m \in M$ with $\|m\|_{1,w} > 2\|x\|_{1,w}$ has $\|x + m\|_{1,w} > \|x\|_{1,w}$. Therefore the infimum may be restricted to the compact set $M \cap \{m : \|m\|_{1,w} \le 2\|x\|_{1,w}\}$ (the intersection of a closed subspace with a compact ball [1, 4]), on which the continuous objective attains its minimum. Definiteness ($\Phi_1([x]) = 0 \iff x \in M_1$), homogeneity, and the triangle inequality follow standard quotient norm properties [1, 4]. The identical argument applies to (C_2, M_2) and Γ_2. □

In words: the leftover is the distance from the vector to the repair subspace, and in finite dimensions a nearest point exists.

Theorem FOC-T2 — full-subspace duality

For a matrix A whose image is the declared repair subspace and positive weights w,

$$\min_y \|x + Ay\|_{1,w} = \max_z \{z^T x : A^T z = 0,\ |z_i| \le w_i\}.$$

Both extrema are attained.

Proof. We formulate the primal quotient minimization as a linear program in standard form. For a vector $x \in C_1$, repair matrix A, and weights $w > 0$, write the residual as $r = x + Ay = r^+ - r^-$ with $r^+, r^- \ge 0$. The primal problem is:

$$\min_{y, r^+, r^-} \sum_i w_i (r_i^+ + r_i^-) \quad \text{subject to} \quad r^+ - r^- - Ay = x, \quad r^+ \ge 0, \quad r^- \ge 0.$$

Assigning an unconstrained Lagrange multiplier vector z to the equality constraint $r^+ - r^- - Ay = x$, the dual objective is $z^T x$. Dual feasibility requires:

1. Non-negativity on the reduced costs for r^+: $w_i - z_i \ge 0 \implies z_i \le w_i$.
2. Non-negativity on the reduced costs for r^-: $w_i + z_i \ge 0 \implies z_i \ge -w_i$, so $|z_i| \le w_i$.

3. Vanishing reduced cost for unconstrained y: $-A^T z = 0 \implies A^T z = 0$. The resulting dual problem is $\max_z\{z^T x : A^T z = 0, |z_i| \le w_i\}$. Since the primal is feasible ($y = 0, r^+ = \max(x, 0), r^- = \max(-x, 0)$) and bounded below by 0, strong linear programming duality [2, 3] ensures that both primal and dual optimal values are attained and strictly equal. □

The dual vector z is a price: it is blind to every licensed repair, and it charges at most the observer weight on each coordinate. Its pairing with x cannot exceed the leftover, and strong duality says some such z achieves equality. The annihilation condition is annihilation of **every vector in** $\operatorname{im} A$, rather than only the original named generators before closure. If a declared operation sends e_1 to e_2, the repair matrix must span both before a functional can certify the quotient.

Corollary FOC-C1 — exact separation

$[x]$ is nonzero in $C/\operatorname{im} A$ if and only if some dual vector z with $A^T z = 0$ and $|z_i| \le w_i$ satisfies $z^T x > 0$.

Proof. If $[x] = 0$, then $x \in \operatorname{im} A$, so $z^T x = 0$ for every z with $A^T z = 0$. If $[x] \neq 0$, Theorem FOC-T2 gives $\max z^T x = \|x\|_{1,w/M} > 0$. Because the maximum is attained, there exists a feasible z^* with $z^{*T} x > 0$. □

This exact separation by an annihilating dual witness reflects classical geometric separation and annihilator duality for closed subspaces in normed linear spaces [1, 2, 4].

Implementer remark — feasible witnesses and exact optima

A single normalized basis vector of $\ker A^T$ is a feasible dual point, whose pairing supplies only a lower bound. For $A = e_1 \in \mathbb{Q}^3$ and $x = e_3$, the feasible vector $z = e_2$ pairs to zero although $\Phi_1([x]) = 1$. For two directed cycles with circulations $+2$ and -2, adding unsigned cycle indicators also gives a zero pairing, while the exact quotient value is 4. Zero at a selected feasible witness therefore does not establish exact repair.

The seven exact conformance cases in `code/tests/test_foc_conformance.py` cover a zero repair column, repeated coordinates, a multidimensional left kernel, opposite circulations, two triangles, and the positive sign of an optimal witness pairing. They check finite examples of FOC-T2 and FOC-C1. The companion's vertex enumeration is a reference for small instances; its worst-case cost is exponential in dimension. Larger instances require an exact linear-programming solver with independently checked primal and dual feasibility and matching objective values.

Definition FOC-D2 — certificate

A certificate records a primal repair, residual, dual annihilator, common value, and input digest. It is valid only if primal feasibility, full-subspace annihilation, capacity bounds, and equality of primal and dual objectives all replay against the same packet.

3. Observer positivity and nullity invariance

Proposition FOC-P1 — nullity invariance

Changing from strictly positive weights w to strictly positive weights w' changes the values of $\Phi_1([x])$ and $\Gamma_2([d_1 x])$, but not whether they are zero.

Proof. In finite dimensions, all norms on Q_1 are equivalent. Since w and w' are strictly positive on every coordinate, there exist constants $0 < c \le C < \infty$ such that $c\|v\|_{1,w} \le \|v\|_{1,w'} \le C\|v\|_{1,w}$ for all $v \in C_1$. Taking infima over M_1 preserves these inequalities: $c\Phi_{1,w}([x]) \le \Phi_{1,w'}([x]) \le C\Phi_{1,w}([x])$. Therefore $\Phi_{1,w'}([x]) = 0 \iff \Phi_{1,w}([x]) = 0$. The same holds for Γ_2 on Q_2. □

The observer sets prices, not zeros. A zero weight is refused because it would allow an infinite coordinate error to pass without penalty.

4. Direct sums

Let $P = (C_0, C_1, C_2, U_2, d_0, d_1, a_2, w_1, w_2)$ and $P' = (C_0', C_1', C_2', U_2', d_0', d_1', a_2', w_1', w_2')$ be two packets. Their direct sum $P \oplus P'$ has carriers $C_i \oplus C_i'$, repair map $d_0 \oplus d_0'$, next-defect map $d_1 \oplus d_1'$, second-stage repair map $a_2 \oplus a_2'$, and concatenated weights (w_1, w_1') and (w_2, w_2').

Theorem FOC-T4 — direct-sum additivity

For $(x, x') \in C_1 \oplus C_1'$,

$$\Phi_{1,P\oplus P'}([(x,x')]) = \Phi_{1,P}([x]) + \Phi_{1,P'}([x']),$$

$$\Gamma_{2,P\oplus P'}([(d_1x, d_1'x')]) = \Gamma_{2,P}([d_1x]) + \Gamma_{2,P'}([d_1'x']).$$

Proof. The objective decomposes: $\|(v,v')\|_{1,(w,w')} = \|v\|_{1,w} + \|v'\|_{1,w'}$. The repair space decomposes: $\operatorname{im}(d_0 \oplus d_0') = \operatorname{im} d_0 \oplus \operatorname{im} d_0'$. Therefore the minimization uncouples into two independent problems:

$$\min_{f,f'} \left(\|x + d_0 f\|_{1,w_1} + \|x' + d_0' f'\|_{1,w_1'}\right) = \min_{f} \|x + d_0 f\|_{1,w_1} + \min_{f'} \|x' + d_0' f'\|_{1,w_1'}.$$

The same argument applies to Γ_2. □

The direct-sum cochain complex and objective decomposition follow standard direct-sum decoupling in homological algebra and convex optimization [3, 5]. Two uncoupled problems have the sum of their individual leftovers.

5. Transport

Let P and P' be packets, and let $T_0 \colon C_0 \to C_0'$, $T_1 \colon C_1 \to C_1'$, $T_2 \colon C_2 \to C_2'$, and $S_2 \colon U_2 \to U_2'$ be linear maps satisfying:

$$T_1 d_0 = d_0' T_0, \qquad T_2 d_1 = d_1' T_1, \qquad T_2 a_2 = a_2' S_2.$$

These commutation relations ensure that T_1 preserves the repair space ($T_1(M_1) \subseteq M_1'$) and T_2 preserves the second-stage repair space ($T_2(M_2) \subseteq M_2'$), forming cochain morphisms whose induced quotient maps satisfy standard operator norm bounds [4, 5]. Let $\|T_1\|_{1,w_1\to w_1'}$ and $\|T_2\|_{1,w_2\to w_2'}$ denote the weighted operator norms:

$$\|T_1\|_{1,w_1\to w_1'} = \max_j \frac{1}{w_{1,j}} \sum_i w_{1,i}' \, |(T_1)_{ij}|.$$

Theorem FOC-T5 — transport bounds

For any defect $x \in C_1$,

$$\Phi_1'([T_1 x]) \le \|T_1\|_{1,w_1 \to w_1'}\, \Phi_1([x]),$$

$$\Gamma_2'([T_2 d_1 x]) \le \|T_2\|_{1,w_2 \to w_2'}\, \Gamma_2([d_1 x]).$$

Proof. Let f^* be an optimal repair for x in P, so that $\Phi_1([x]) = \|x + d_0 f^*\|_{1,w_1}$. Set $f' = T_0 f^*$. Then $T_1 x + d_0' f' = T_1 x + d_0' T_0 f^* = T_1(x + d_0 f^*)$. Since f' is a feasible repair in P',

$$\Phi_1'([T_1 x]) \le \|T_1 x + d_0' f'\|_{1,w_1'} = \|T_1(x + d_0 f^*)\|_{1,w_1'} \le \|T_1\|_{1,w_1 \to w_1'}\, \|x + d_0 f^*\|_{1,w_1} = \|T_1\|_{1,w_1 \to w_1'}\, \Phi_1([x]).$$

The same argument applies to Γ_2, using $u' = S_2 u^*$ with an optimal second-stage repair u^*. □

A compatible packet map bounds the transported leftover by its operator norm times the original value. Growth alone is compatible with this bound: twice the identity has norm two and doubles the quotient distance. Only a violation of the stated norm bound contradicts these hypotheses.

6. Diagnosis, refusal, and authorization

The pair $(\Phi_1([x]), \Gamma_2([d_1 x]))$ gives three mathematical labels:

1. **ExactRepair:** $\Phi_1([x]) = 0$. The vector belongs to the repair subspace; its leftover is zero.
2. **CoherentResidual:** $\Phi_1([x]) > 0$ and $\Gamma_2([d_1 x]) = 0$. The leftover is nonzero, but its boundary in the next quotient vanishes.
3. **NonclosedDefect:** $\Phi_1([x]) > 0$ and $\Gamma_2([d_1 x]) > 0$. The leftover is nonzero and produces an uncancelled defect one degree up.

Theorem FOC-T3 — diagnostic trichotomy

Every well-posed defect in a valid packet falls into exactly one of these three categories.

Proof. By Theorem FOC-T1, $\Phi_1([x]) \ge 0$ and $\Gamma_2([d_1 x]) \ge 0$. The outcome $\Phi_1([x]) = 0$ and $\Gamma_2([d_1 x]) > 0$ cannot occur: if $\Phi_1([x]) = 0$, then $x \in \operatorname{im} d_0$, so $x = d_0 f$ for some f. Then $d_1 x = d_1 d_0 f = 0$ by the cochain condition. Setting $u = 0$ in the definition of Γ_2 gives $\Gamma_2([0]) \le \|0 + a_2 0\|_{1,w_2} = 0$. Since $\Gamma_2 \ge 0$, $\Gamma_2([d_1 x]) = 0$. The three cases partition the attainable value pairs and therefore classify every valid defect uniquely. □

A diagnostic label is not an authorization. In particular, a coherent residual is a vector whose next defect vanishes in the quotient. It is not an assertion that the vector is desirable, stable, or permitted to act. The mathematical diagnostic labels (`ExactRepair`, `CoherentResidual`, `NonclosedDefect`) strictly classify the cochain vector; operational authorization and descent decisions require explicit grounded gates governed by CODA-FDA [9].

The reference companion supports a bounded stabilization check for a declared finite registry of exact additive corrections. The caller supplies the same packet, defect, scope, and complete finite list independently of the evidence record. A correction is improving when it strictly lowers Φ_1. Each correction is replayed with the exact quotient solver; the packet and observer are frozen. The record binds the

concrete corrections, every replay result, evaluator version, scope, and budget. The budget must cover every listed move.

`StabilizedSurvivor` means that a coherent residual has no improving correction in that supplied finite list. Completeness is relative to the declared list; this implementation does not establish completeness of an external licensing policy, search over undeclared operators, authenticate an issuer, or grant permission. Digest strings without the matching registry and successful replay leave the route `OpenObstruction`. Invalid algebraic input raises a refusal exception before any mathematical diagnostic is returned.

OpenObstruction is an operational evidence-state label. It is not the mathematical diagnostic NonclosedDefect.

No finite iteration bound is asserted here. In particular, a budget counter does not bound a height that later operations may increase.

7. Computational companion

The production module `code/finite_obstruction.py` validates packets, solves the four primal and dual linear programs needed for Φ_1 and Γ_2, checks the full-subspace annihilator equations, returns digest-bound certificates and the total diagnostic, and applies the optional later record of Section 6. A survivor route requires `FiniteRepairRegistry` replay, including all concrete corrections and their exact outcomes; syntactically valid hashes or caller-written dispositions are insufficient.

Adversarial tests include:

- the ℓ^1 versus ℓ^2 instance $A = (1, 1, 1)^T$, $x = (0, 0, 10)$;
- refusal of zero observer weights;
- a generator-closure counterexample where e_1 reaches e_2 and $z = (0, 1)$ is not an annihilator of the full repair subspace;
- exact, coherent-residual, and nonclosed-defect packets;
- direct sums, transport squares, and partial later records.

Exact rational solving via `LinearRepairProblem` supplies algebraic certificates without floating-point error. Lean 4 files in `lean4/` [7] formalize stated logical and algebraic kernels, with no `sorry` stubs:

- `TerminalCompleteness.lean` — abstract decidable trichotomy partition underlying Theorem FOC-T3;
- `L2OAKernel.lean` — logical separation implications of Theorem FOC-T3 conditional on four declared quotient-kernel facts;
- `Algebra.lean` (`FOC.lean`) — dual separation theorem `Coda.FOC.separator_exists` (Corollary FOC-C1) and basic quotient algebra;
- `EpkPrimitiveDifference.lean` and `PrimitiveDifference.lean` — quotient factorization of local diagnostics;
- `Core.lean` — re-exports the files above.

The general theorem follows from finite LP strong duality; each computed rational instance carries replayable exact primal and dual witnesses with equal objectives.

8. Limitations

The calculus is linear and finite-dimensional. It does not treat nonlinear repair maps. The Lean files formalize the abstract decidable trichotomy partition and conditional logical/algebraic kernels; Python exact rational certificates verify individual problem instances; and the general numerical LP duality theorem is established by the manuscript's mathematical proof. They do not formalize simplex-method polyhedral duality in Lean.

Appendix: Context within CODA Architecture

CODA stands for the Calculus of Obstruction, Descent, and Authorization. It is an exact, finite mathematical apparatus that enforces strict claim discipline by separating linear repairs, transport certificates, and finite descent from ungrounded authority claims. The architecture was established in the frozen baseline derivation [11]. The obstruction calculus presented here (FOC) interacts with adjoining layers, notably the typed residual registers of CODA-A [8], the descent and authorization gates of CODA-FDA [9], and representation transport in CODA-MUR [10]. The system comprises eight operational layers governed by a single overarching authority architecture (GTLA).

Layer	Abbreviation	Layer / Component Name & Scope
1	**FOC**	Finite Obstruction Calculus
2	**MUR**	Meta & Universal Re-Presentation (what presentations exist)
3	**CS**	Continuum Shadow (what continuum presentation may descend to a finite calculus system)
4	**CS_Q**	Continuum Shadow to Continuum Bridge (how continuum presentations descend)
5	**Q**	The Continuum (continuum presentations via completion and enclosure)
6	**A**	Anomalon (global residuals not removed and typed instead)
7	**OR**	Omniversal Re-Presentation (what presentations can transport)
8	**FDA**	Finite Descent Authority (what presentation may descend to evaluation and authorization)
9	**GTLA**	Layered Authority Architecture for CODA

No theorem or certificate in this paper depends on the other CODA layers; the obstruction calculus presented here is self-contained.

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